Wireless Digital Twin Calibration: Refining DFT-Domain Channel Information

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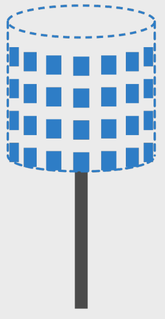
Asilomar Conference on Signals, Systems, and Computers, 2026

Wireless Intelligence Lab

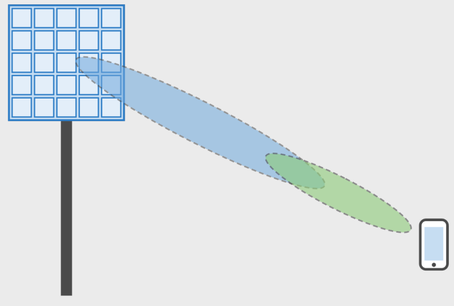
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MIMO matters in 6G and beyond

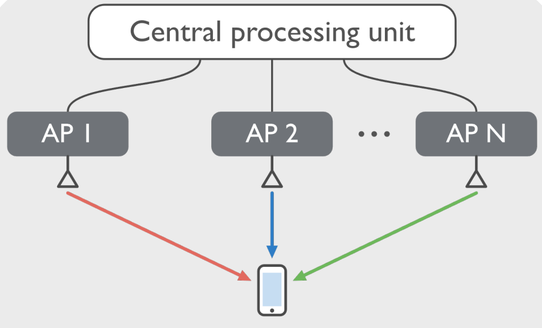
- Large-scale MIMO is a key enabler for next-generation wireless systems



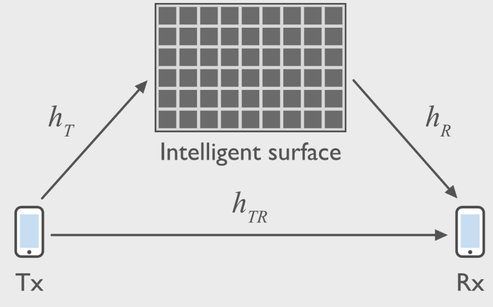
Massive MIMO



mmWave and THz MIMO



Cell-free massive MIMO



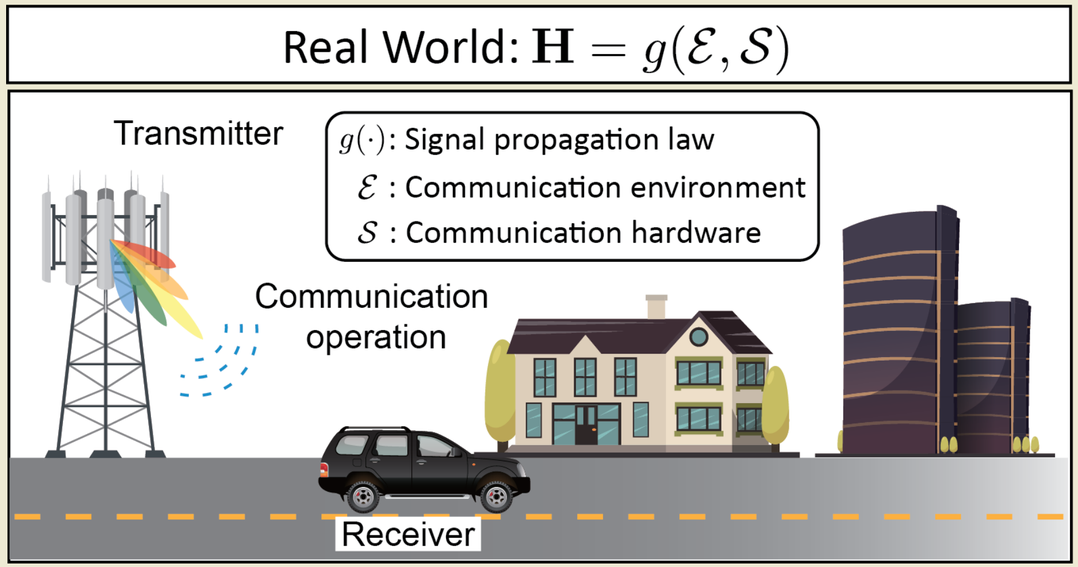
Intelligent surfaces

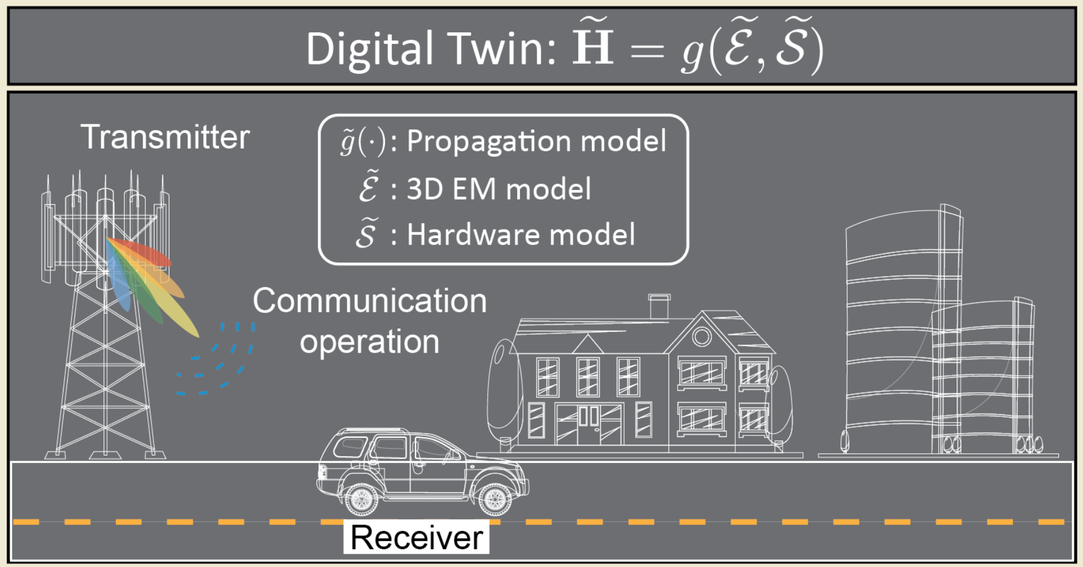
- However, the channel acquisition overhead grows with the number of antennas
  - Pilot/beam training overhead
  - Feedback overhead
  - Control signaling overhead



Channel acquisition overhead hinders large-scale MIMO from scaling gains

Wireless digital twins





- Real-world channels are determined by
  - Communication environment
  - Communication hardware
  - Signal propagation laws
- Digital replica approximates real world with
  - 3D EM environment model
  - Communication hardware model
  - Signal propagation model

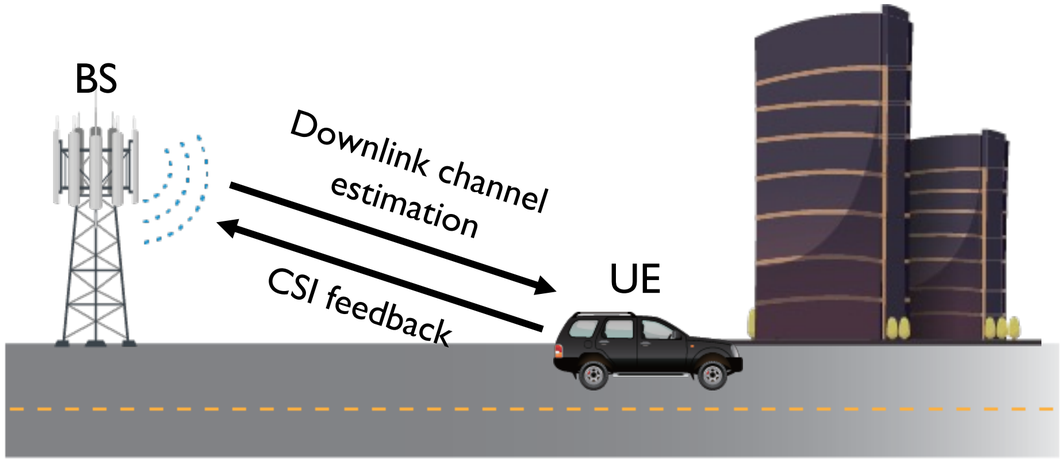
Wireless digital twins can generate channel priors to help real-world communications

[Alkhateeb'23] A. Alkhateeb, S. Jiang and G. Charan, "Real-Time Digital Twins: Vision and Research Directions for 6G and Beyond," IEEE Communications Magazine

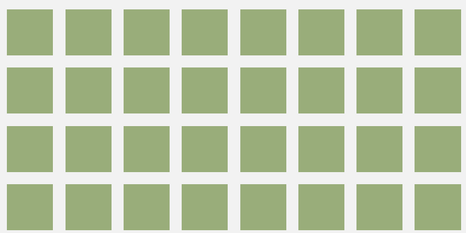
Digital twin aided CSI acquisition

Codebook-based CSI feedback in 3GPP

Goal of CSI feedback:
BS learns the dominant multi-path components



Conventional channel estimation needs

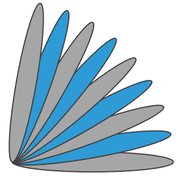


≥ **32**
orthogonal pilots
for 32 BS antennas

1. BS sends CSI-RS pilots; UE estimates the full channel

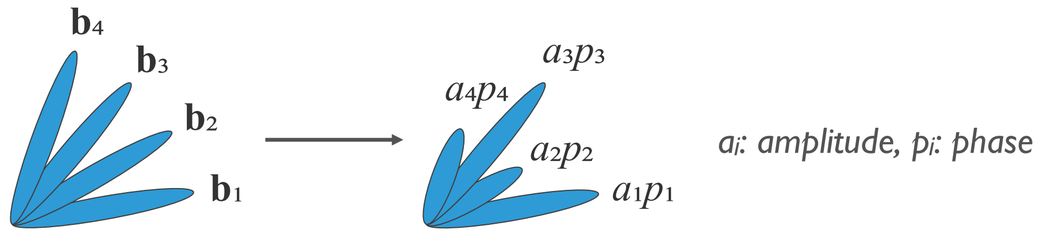
$$\mathbf{h} = \sum_{i=1}^{L} a_i \mathbf{a}(\theta_i, \phi_i)$$

2. UE projects the channel onto the DFT codebook



orthogonal DFT beams (3GPP Type-II)

3. UE picks the strongest beams and their weights



a_i: amplitude, p_i: phase

4. UE reports them; BS reconstructs the channel

$$\mathbf{f} = \sum_{i=1}^{4} a_i p_i \mathbf{b}_i$$

beam indices + quantized weights

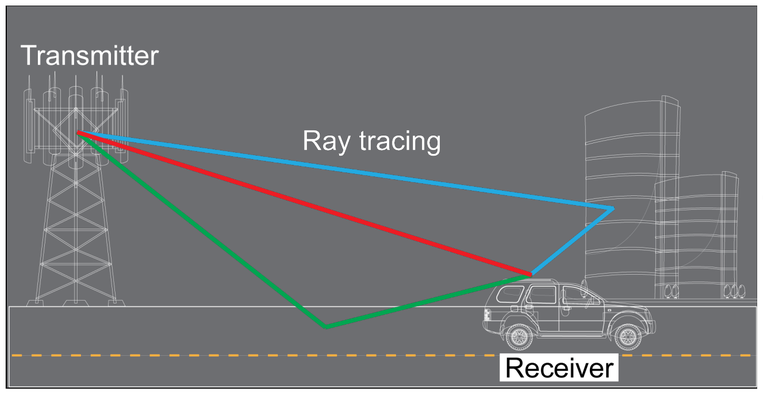
CSI-RS pilots grow with the number of BS antennas

[3GPP] 3GPP TS 38.214, "Physical Layer Procedures for Data (Release 15)," v15.16.0, 2022.

Digital twin aided CSI acquisition

Digital Twin

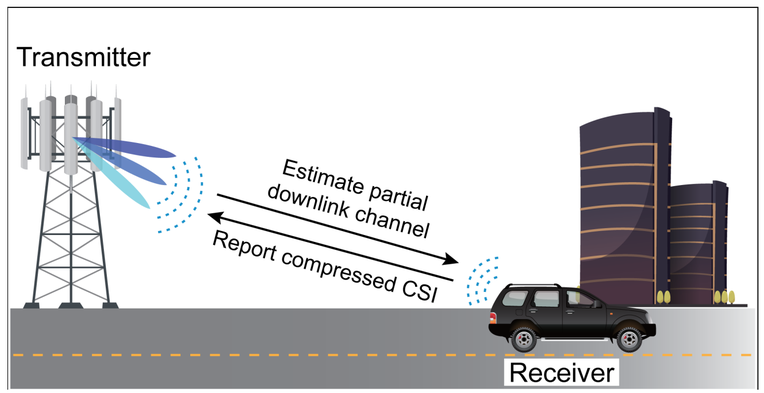
1 BS runs ray tracing in its digital twin



→ *synthetic channel info.: dominant beams*

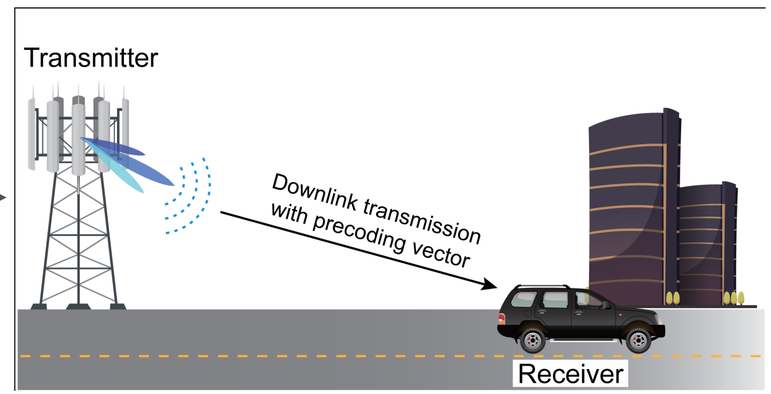
DT-aided codebook-based CSI feedback

2 BS sends pilots only on the predicted beams



→ *UE estimates only a P-dim channel*

3 BS precodes the downlink with the reported CSI

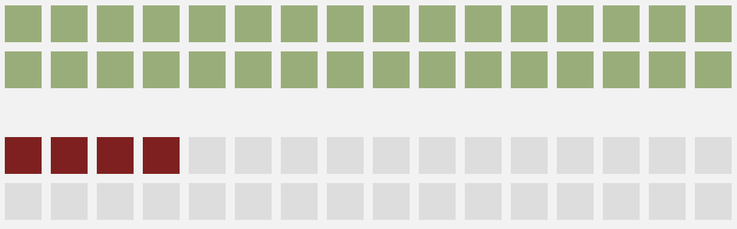


→ *no full-channel estimation at the UE*

Downlink signaling overhead can be reduced

Conventional
32 pilots

DT-aided
P = 4 pilots



P = 4 is an example, using 4 DFT beams to represent the multi-path components

Complexity at the UE side can be reduced

Effective channel of DFT beam

Pilot

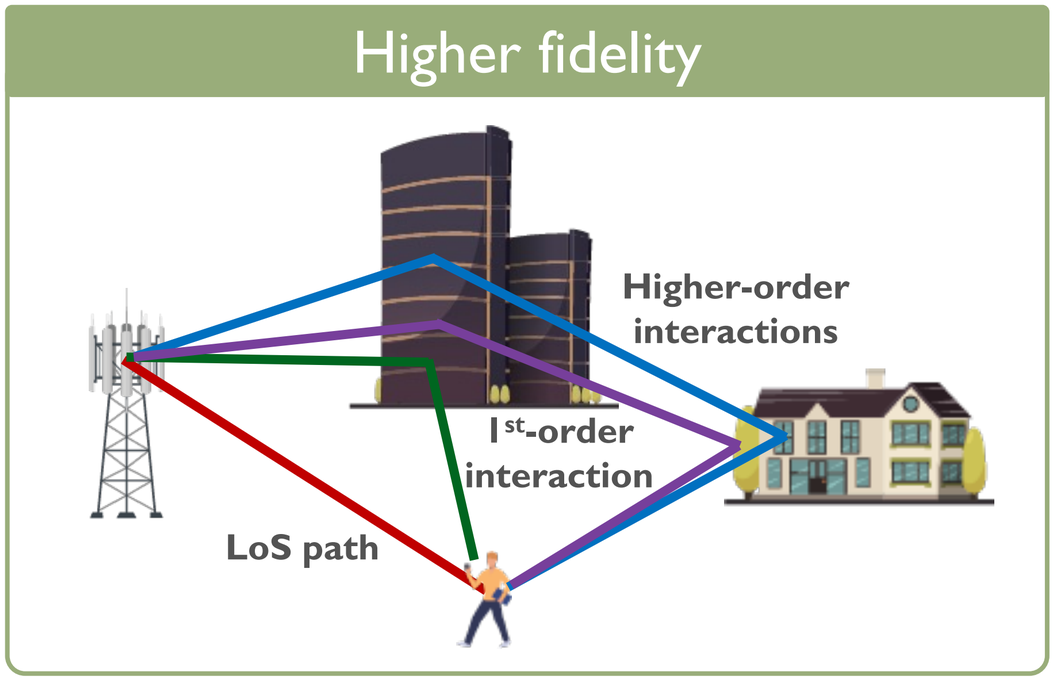
$$y = \mathbf{h}^H \mathbf{f} s + n$$

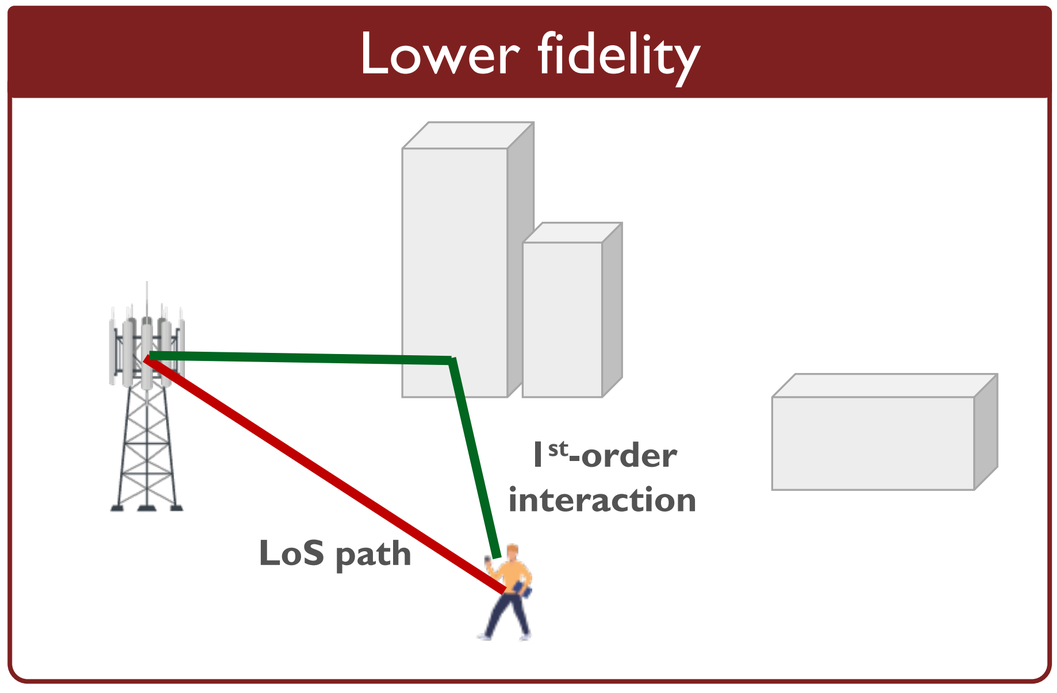
Digital twins cut the pilot overhead by predicting the dominant beams

Accuracy–complexity trade-off in digital twins

However, the computational complexity of digital twin is crucial for real-time operation

- High-fidelity digital twins provide accurate channels, but ray tracing is time-consuming
- Low-complexity digital twins are fast, but may miss multi-path components





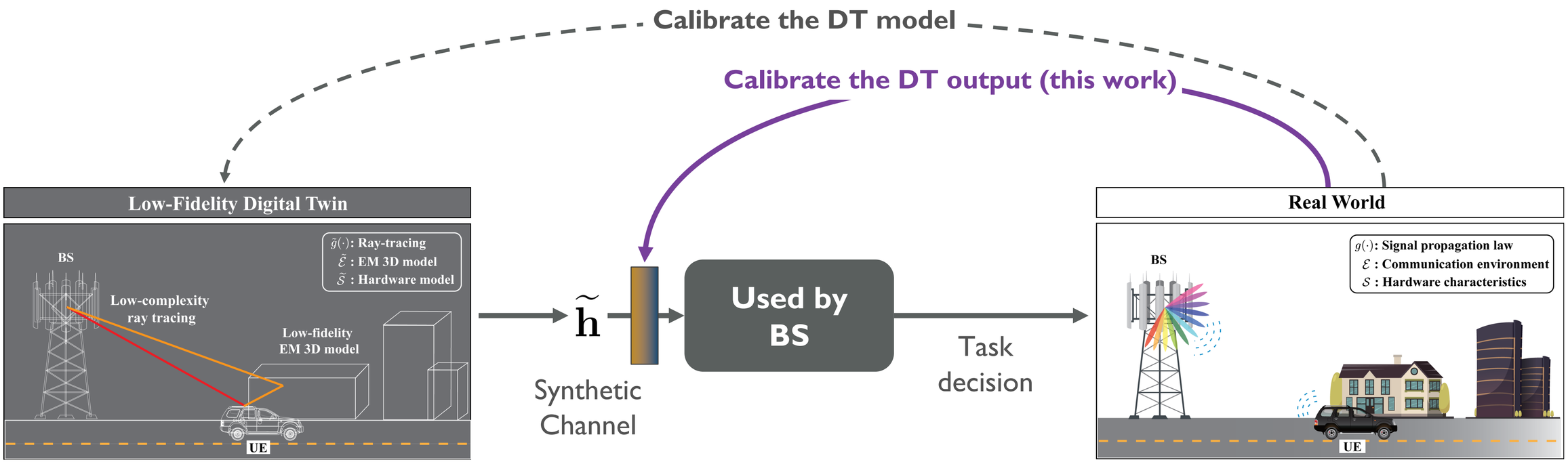
For CSI feedback, we need low-complexity digital twins

Can we calibrate a low-complexity digital twin to approach high fidelity?

[Luo'26] H. Luo, M. Gupta, T. A. Khan and A. Alkhateeb, "Low-Complexity Digital Twin for CSI Acquisition in MIMO Communications," IEEE ICC, 2026.

Calibration motivation and approaches

- Approaches: calibrate the digital twin output



- Motivation
  - Obtain more accurate digital twin predictions
  - Reduce complexity and latency: low-fidelity digital twin + lightweight calibration ML model

This work: calibrate the output of a low-complexity digital twin

Calibrate digital twin output

Problem formulation: Refining DFT-domain channel information

- DT output: DFT weights of the DT channel

$$\widetilde{\mathbf{z}}_k = |\mathbf{D}^H \widetilde{\mathbf{h}}_k| \in \mathbb{R}^N, \quad \mathbf{D} = [\mathbf{d}_1, \ldots, \mathbf{d}_N]$$

DFT codebook with N beams

- Calibration inputs
  - User position $\mathbf{p}_k \in \mathbb{R}^3$
  - low-fidelity DFT weights $\widetilde{\mathbf{z}}_k$

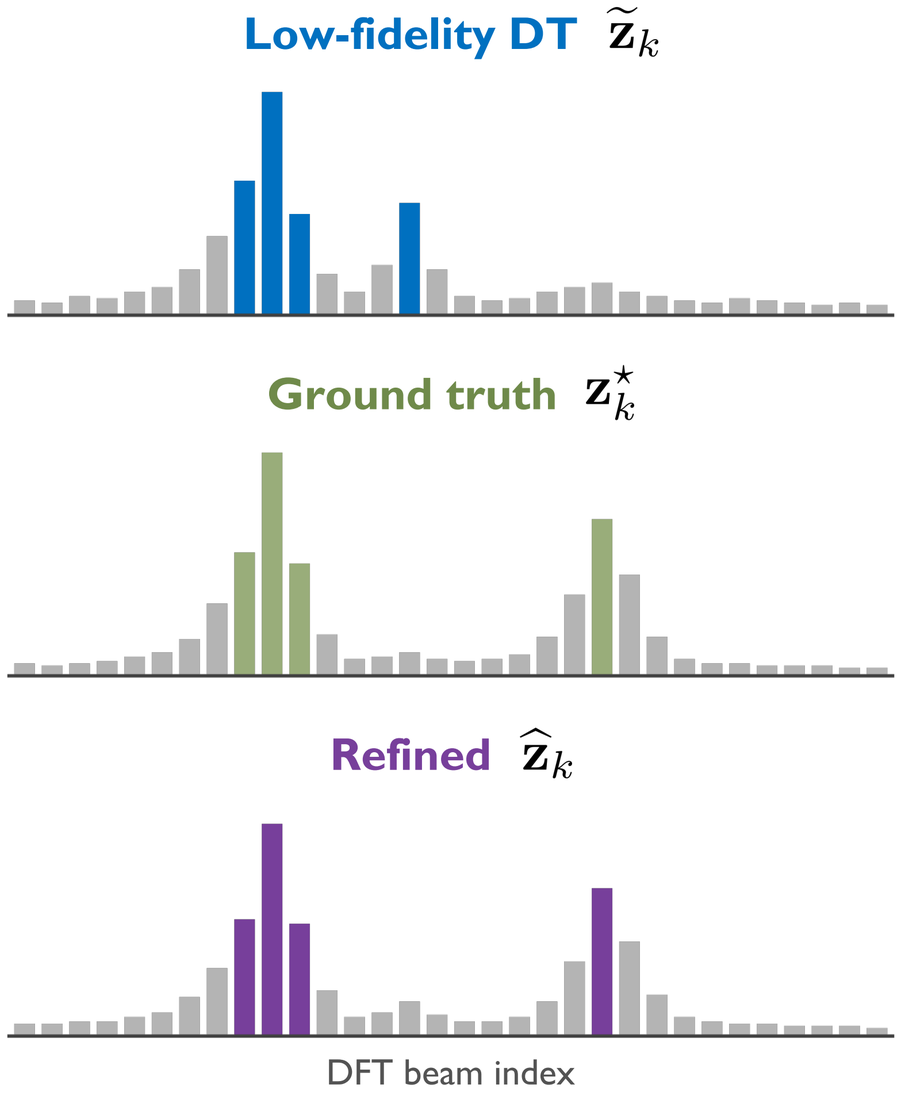
- Optimization problem

$$\min_{\Theta} \quad \sum_{k=1}^{K} \|\mathbf{z}_k^\star - \widehat{\mathbf{z}}_k\|_2$$

Ground-truth DFT weights

$$\text{s.t.} \quad \widehat{\mathbf{z}}_k = f(\mathbf{p}_k, \widetilde{\mathbf{z}}_k; \Theta), \ \forall k$$

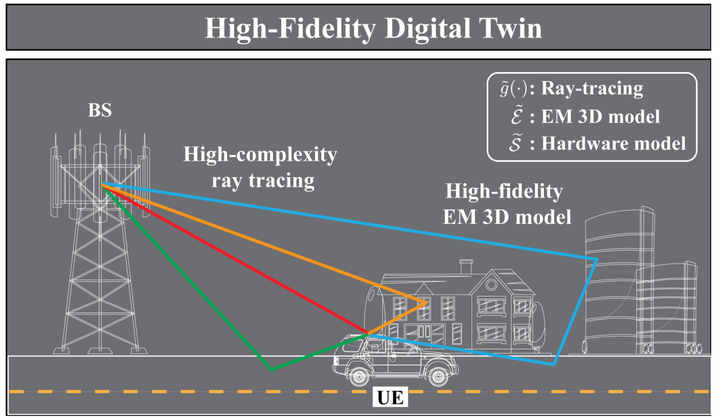
Calibration model with learnable parameters Θ



User position helps resolve ambiguities in DFT weights, e.g., for NLoS users

Proposed solution: Digital twin calibration framework

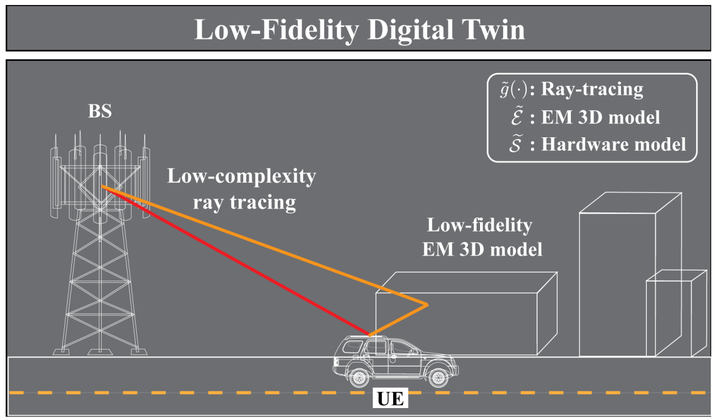
- Key idea: Calibrate the DFT-domain output of a low-complexity digital twin



High-fidelity DT channel

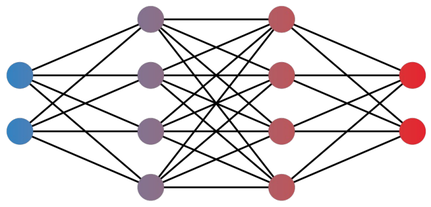
Offline training data

- Paired low- and high-fidelity DT weights
- Labels: full DFT weights of the high-fidelity channel



$\widetilde{\mathbf{z}}_k$

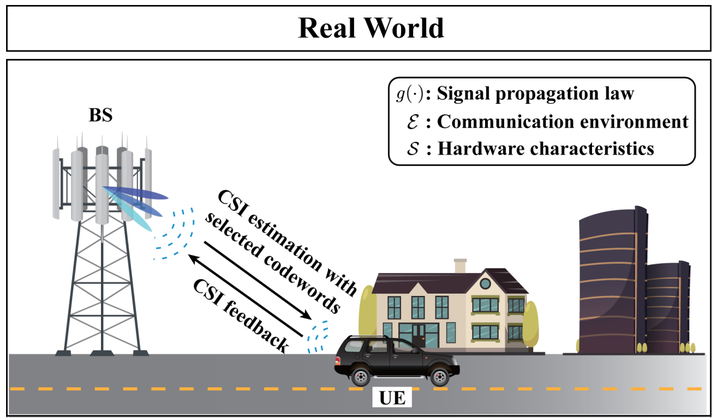
Low-fidelity DFT weights



Online refinement with deep learning

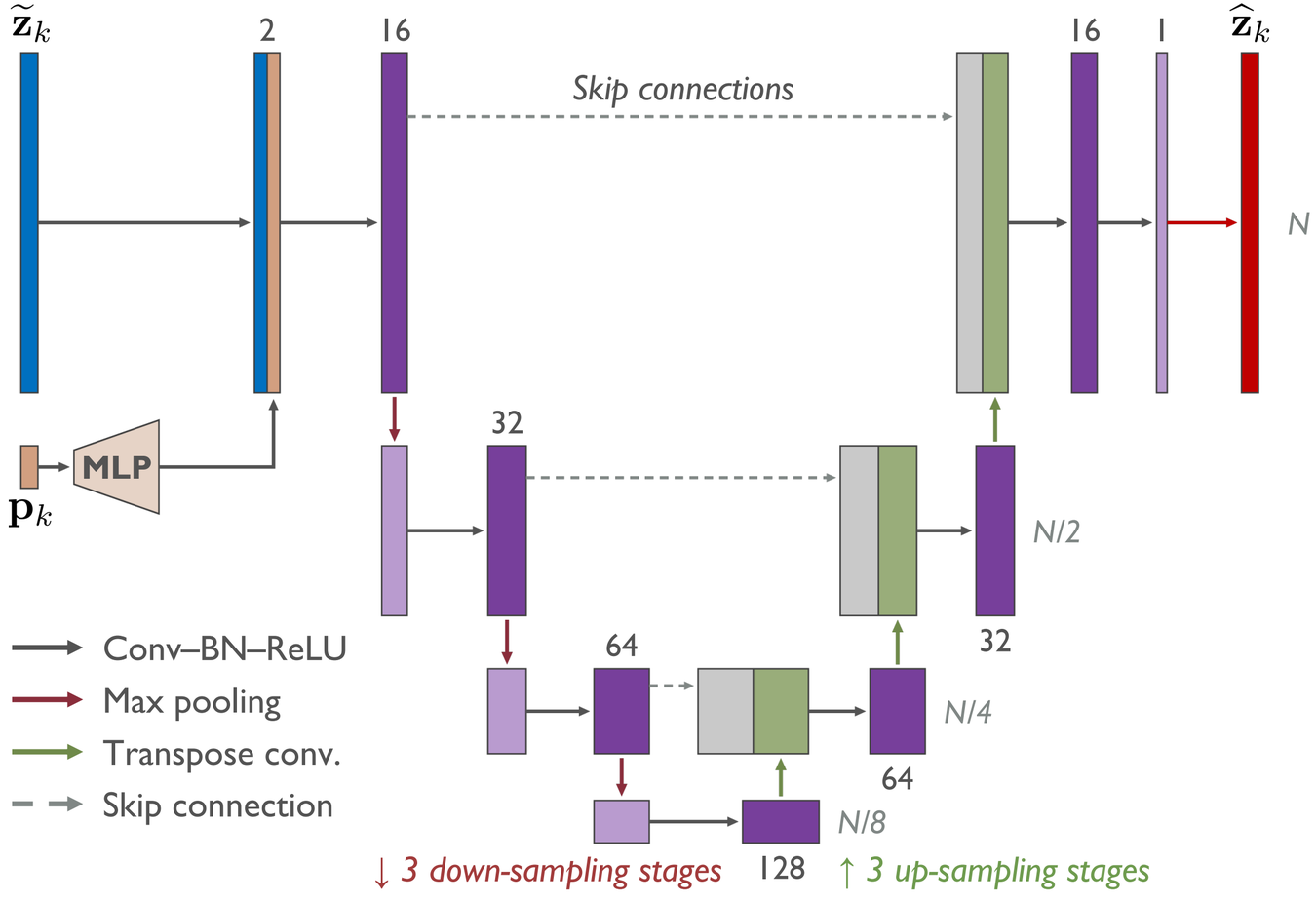
$\widehat{\mathbf{z}}_k$

Refined DFT weights



Train offline with a high-fidelity DT; run lightweight calibration online

Proposed solution: Lightweight 1D U-Net calibration model



- Why a U-Net with skip connections?
  - Low-fidelity input is well structured
  - Skip connections keep this structure
  - The network learns only corrections
- Loss function (MSE)

$$\mathcal{L}(\mathbf{z}_k^\star, \widehat{\mathbf{z}}_k) = \|\mathbf{z}_k^\star - \widehat{\mathbf{z}}_k\|_2^2$$

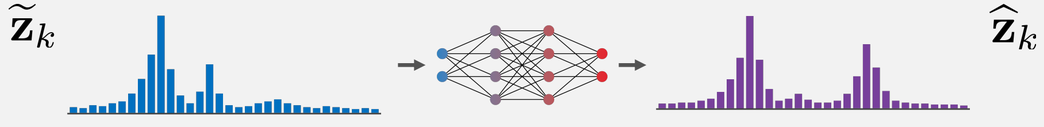
$\mathbf{z}_k^\star$: Ground-truth DFT weights; $\widehat{\mathbf{z}}_k$: Refined DFT weights

Lightweight model with only 181K parameters to refine the DFT weights

[Ronneberger'15] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in Proc. of MICCAI, 2015, pp. 234–241.

Case study: Digital twin aided codebook-based CSI feedback

1 Calibrate the DFT weights from the digital twin

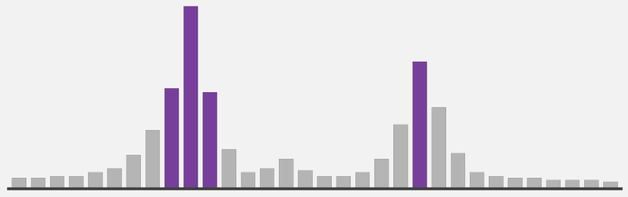


low-fidelity → refined DFT weights

$$\widehat{\mathbf{z}}_k = f(\mathbf{p}_k, \widetilde{\mathbf{z}}_k; \Theta)$$

refined weights — low-fidelity weights

2 Select the top-*P* DFT codewords

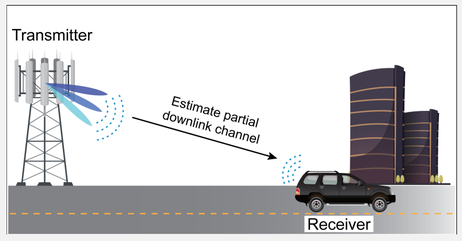


Top-P beams
(P = 4 as an example)

$$\{\widehat{i}_{k,1}, \ldots, \widehat{i}_{k,P}\} = \underset{i \in \{1,\ldots,N\}}{\arg\,\mathrm{sort}} \left[\widehat{\mathbf{z}}_k\right]_i$$

selected beam indices — refined weights

3 Send pilots precoded by the selected codewords

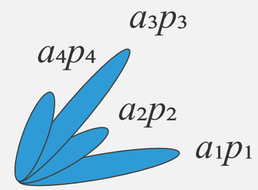


Pilots per user: N = 32 → P = 4
(P = 4 is an example)
Same feedback overhead

$$\widehat{\mathbf{y}}_k = \mathbf{h}_k^H \widehat{\mathbf{Q}}_k \mathbf{S}_k + \mathbf{n}_k$$

selected codewords — pilots

4 User reports coefficients; BS reconstructs CSI



UE reports P beam weights;
BS combines the beams

$$\widehat{\mathbf{w}}_k = \widehat{\mathbf{Q}}_k \widehat{\mathbf{x}}_k$$

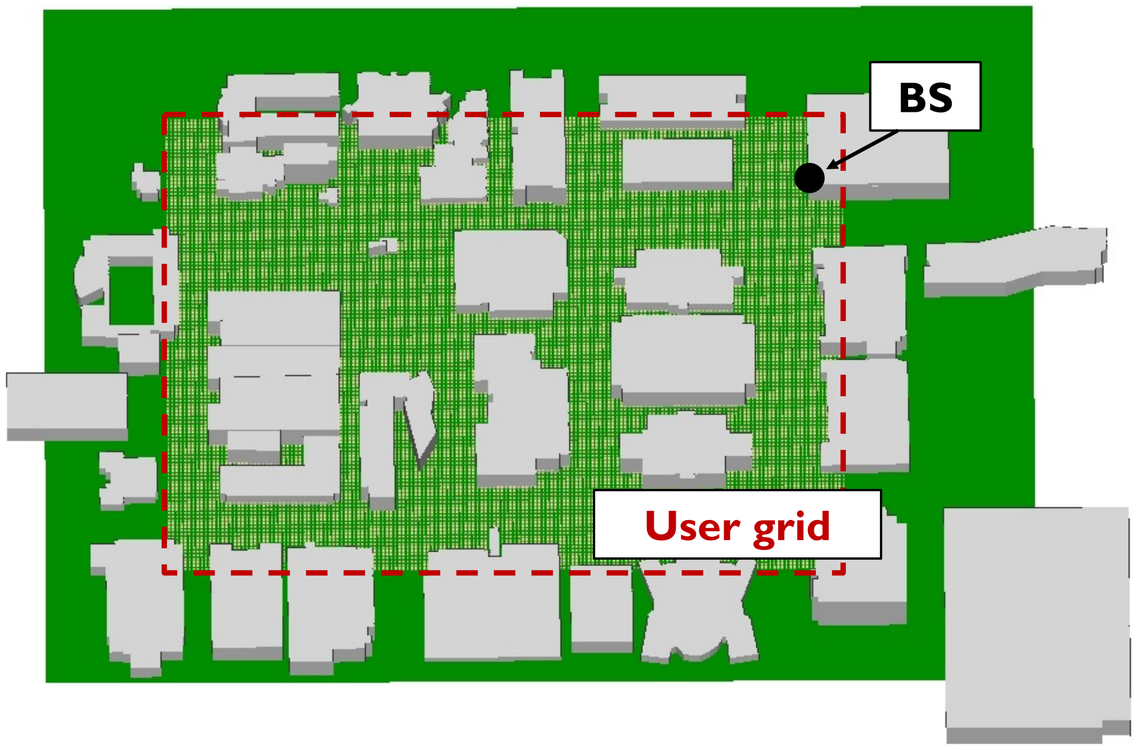
reconstructed CSI — reported coefficients

Refined DFT weights identify the most relevant codewords before CSI feedback

Simulation results

Simulation setup

- Target scenario (high-fidelity DT)
  - Wireless InSite: 6 reflections, 1 diffraction, diffuse scattering
- Baseline scenario (low-fidelity DT)
  - Sionna RT: 6 interactions, 1st-order diffraction model, diffuse scattering
- System parameters
  - 3.5 GHz, 32-antenna ULA at a height of 22 m
  - DFT codebook with 32 codewords, $P = 4$
- Dataset (DeepMIMO channel generator)
  - User grid: 410 m × 320 m, 2.5 m spacing
  - 13,191 training and 7,895 off-grid test samples

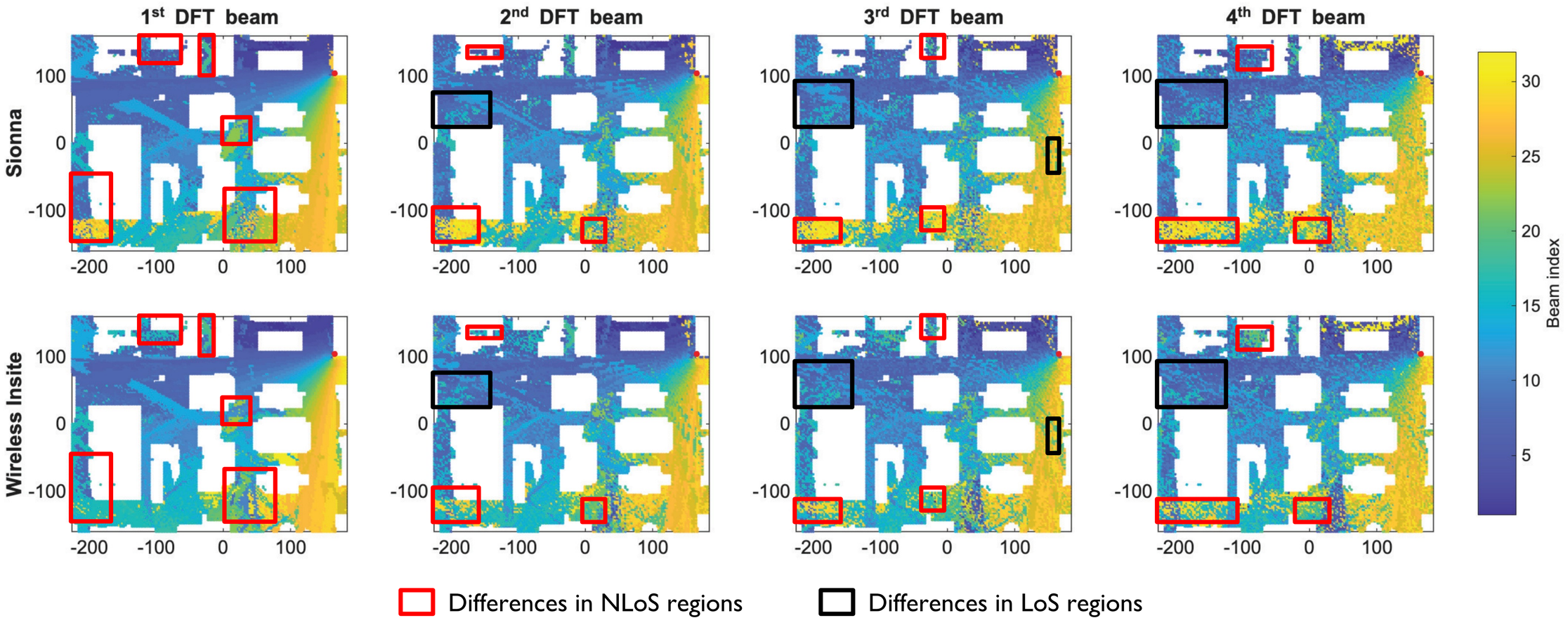


ASU campus (bird's-eye view)

[Remcom] Wireless InSite, http://www.remcom.com/wireless-insite
[Hoydis'23] J. Hoydis et al., "Sionna RT: Differentiable Ray Tracing for Radio Propagation Modeling," in Proc. of IEEE Globecom Workshops, 2023.
[Alkhateeb'19] A. Alkhateeb, "DeepMIMO: A Generic Deep Learning Dataset for Millimeter Wave and Massive MIMO Applications," in Proc. of ITA, 2019.

What is the impact of a low-complexity digital twin?

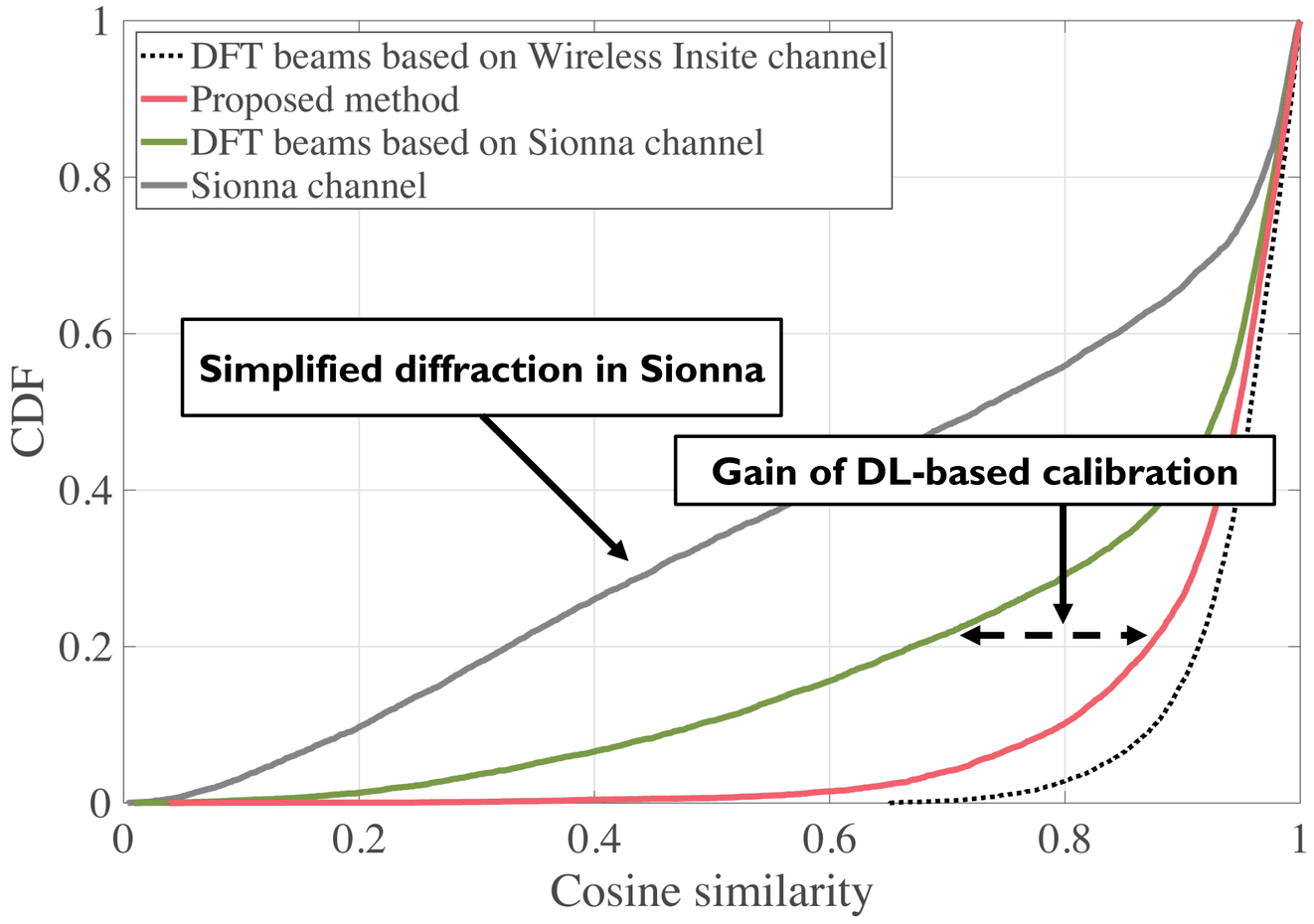
- Top-4 DFT beam indices: Sionna RT (low fidelity) vs. Wireless InSite (high fidelity)



Strongest beam differs mainly in NLoS; other beams differ in both LoS and NLoS

These differences result from the simplified diffraction assumption in Sionna RT

How effective is the proposed calibration approach?



- Metric: cosine similarity

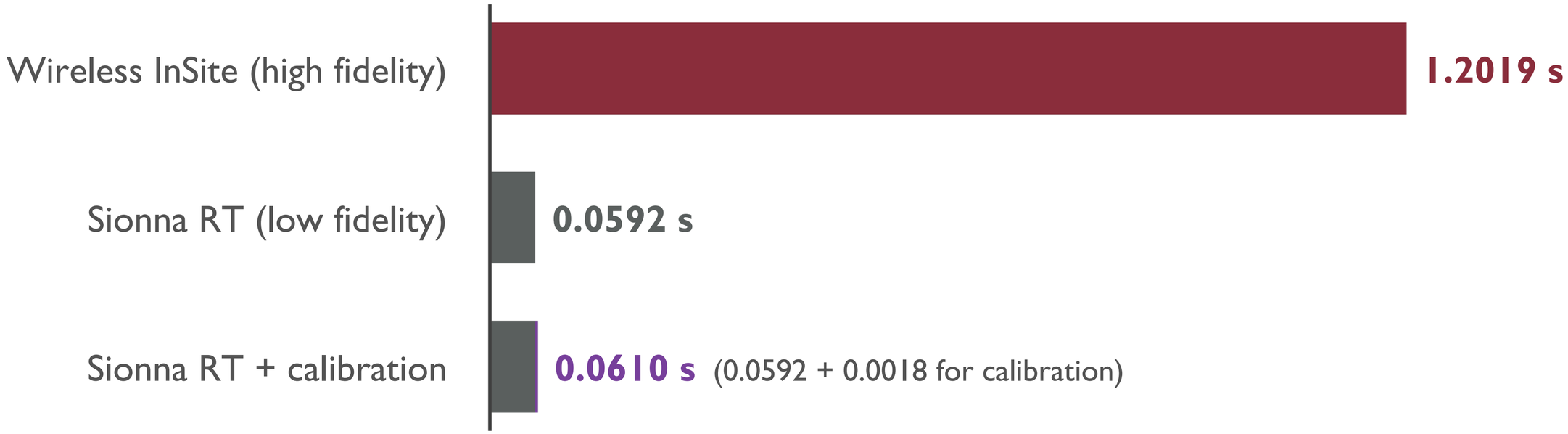
$$\rho(\mathbf{h}, \widehat{\mathbf{w}}) = \frac{|\mathbf{h}^H \widehat{\mathbf{w}}|}{\|\mathbf{h}\|_2 \, \|\widehat{\mathbf{w}}\|_2}$$

- Benchmarks (top-4 DFT beams)
  - Upper bound: Wireless InSite CSI
  - Lower bound: Sionna channel
- Raw Sionna channel performs poorly

The proposed method approaches the high-fidelity upper bound

How much computational overhead can be reduced?

- Computation time per sample (NVIDIA RTX A5000 GPU)



~20×
lower computation time

181K
model parameters

Near high-fidelity accuracy with a low-complexity digital twin

Conclusion and future work

- We propose a digital twin calibration framework for DFT-domain channel information
  - A lightweight U-Net refines low-fidelity DFT weights using the user position
  - Trained offline with high-fidelity DT data

- Key findings
  - A low-complexity DT degrades CSI accuracy, mainly for NLoS users
  - Calibration approaches the high-fidelity DT accuracy
  - Computation time is reduced from 1.2019 to 0.0610 s/sample (~20×)

- Future work
  - Adapting the DT to dynamic, real-world environments
  - Extending the calibration to beam management problem

Q&A

Thank you!

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